

Sensors and the electronics function with very high accuracy, capturing high degree of details, including your foot landing, roll and push-off. The electronics are designed such that these last about a week on their charge. It takes about 15-20 minutes to charge each insole to its full level.

The software

The software and analytics are built on the fundamental philosophy of providing real-time, actionable guidance. This requires enormous effort in ensuring that the models based on which the analytics are done, can calculate with just enough accuracy and return the data to the app real-time, while being able to handle network latency and connectivity gaps.

The challenges

Being placed in a person's shoe and expecting the device to function while the person is running causes the device to experience high amounts of stress on its components. The team explained that they had to do the industrial design in such a way that, while critical components are designed to be kept out of the way of stress from the device, sensors are still able to collect data from points where there is maximum pressure, while simultaneously ensuring that it does not affect other components and are shielded from pressure.

An important design challenge of this device was the existence of two wearable devices working in tandem—one sole on each foot. On the Cloud side, they needed to ensure a regular sync mechanism with the app, and at the same time ensure that all calculations on the app side get done within one or two seconds, so that information and guidance can be communicated to the user in real-time.

"Our end-goal was to design a product for runners, which is extremely non-intrusive and has

Spark of innovation

We enjoy running very much, and like all runners know, running form and style are extremely important in order to continue to enjoy running and improve our runs. When we checked, however, we found out that all currently available fitness trackers and devices are essentially glorified pedometers. These do not help with the fundamentals of running form, understanding the uniqueness of a person's body or providing actionable guidance.

—Anshuman Singh and Shuvadeep Sarkar, co-founders



Stridalyzer

a long battery life, but still provides the software component of the product with all the data that is needed to understand a runner's form. For this purpose, we iterated over various existing solutions and finally had to invent our own solution. We have applied for a patent for it."

On the software side, the team explained that defining the user-experience, so that the user could run without too many interruptions during a run, but get important information at the same time, was challenging. The design team worked with coaches, UI design experts and others to come up with an optimal solution.

How unique is it

There is currently no product in the market that does the same thing; closest products are Sensoria's socks and runScribe. However, these devices only work on one foot and are comparatively limited in their guidance and injury-prevention advice.

The team has filed for a patent for the hardware design, architecture and the way data is collected.

Slapping your feet on the pavement to push forward is not the best way to run

The innovation also includes the methodology to objectively decide optimal sensor placement for running as well as various other sports. "Our unique hardware architecture enables us to provide users with a battery life that is multiple times that of competitive products," they claim.

The product was first launched on Kickstarter in December 2014. It was priced at US\$ 90, without ExpertConnect add-on service or shipping costs.

Covered by over half-a-dozen in the first few weeks of its listing, this device could be a boon to every runner who understands the importance of correct training. In the words of Lynn Jennings, the first American woman to win a long-distance track medal in the Olympics, "If I am not out there training, someone else is." ●